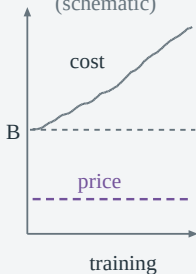


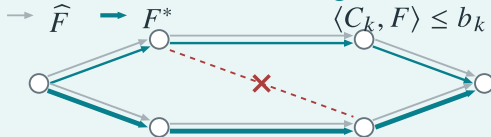
Fixed penalty

Budget leakage
(schematic)



SafeOT

One flow, all budgets



-X- hard constraint: excluded edge
nodes: clustered states · edges: transition flow

Entropic projection

λ_k^*

Can price before violation

clip + EMA $\rightarrow \bar{\lambda}_k^*$

One price rule

feedback $\bar{J}_k - B_k \rightarrow \lambda_k = \bar{\lambda}_k^* + \beta_k \xrightarrow{\tilde{A}} \text{PPO/TRPO} \rightarrow \pi_{t+1}$

Lagrangian

Late feedback
(schematic)



Exact settings

Conditional limits

Fixed penalty

$c = 0, \beta_k$ fixed

Lagrangian

$c = 0, \beta_k$ adaptive

SafeOT

finite ε

LP

$\varepsilon \rightarrow 0$

Active-set

$\varepsilon \rightarrow \infty$
saturated clip